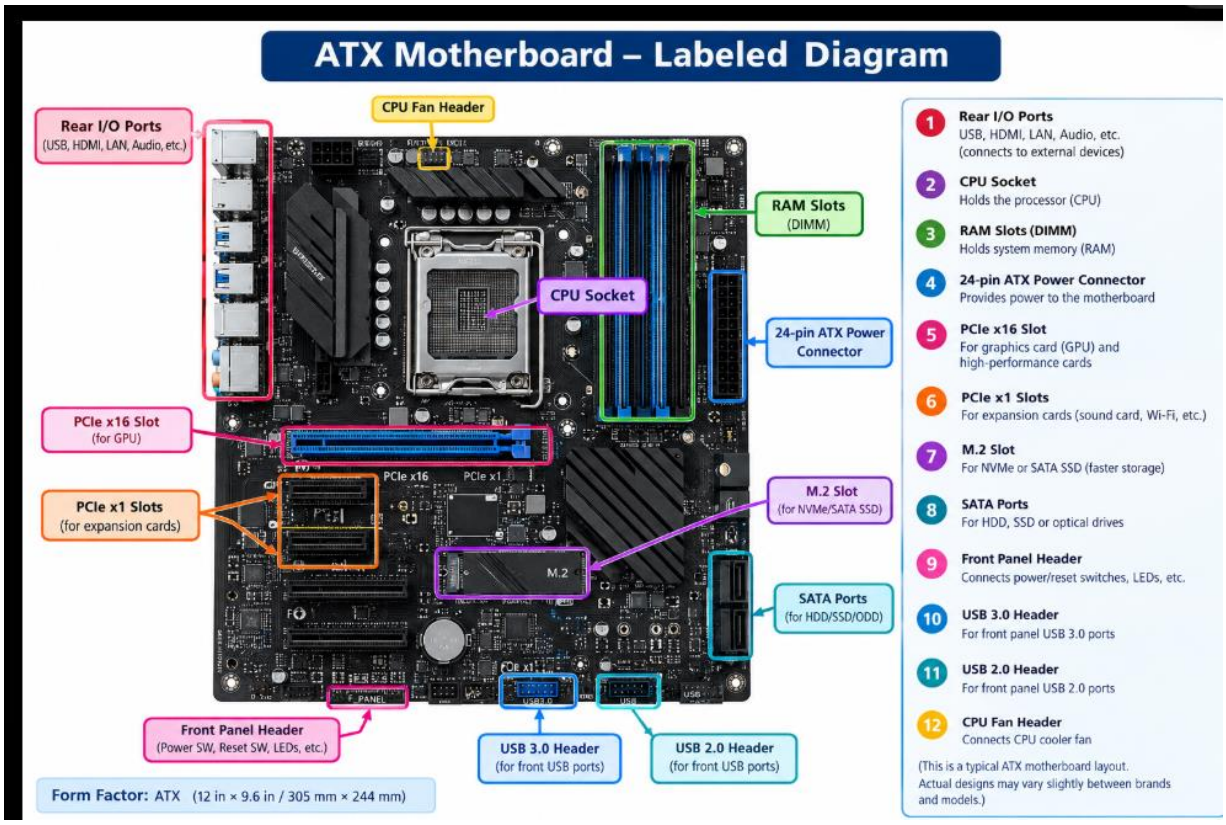
Motherboard and CPU theory

1.
Draw a labeled diagram of an ATX motherboard and mark at least 8 different slots and ports (such as RAM slots, PCIe slots, SATA ports, USB headers, CPU socket, power connector, etc.).



ATX Motherboard – Labeled Diagram

The following diagram shows an ATX motherboard with more than 8 important slots and ports.

Labels and Functions of Motherboard Components

No.	Component Name	Function
1	CPU Socket	Holds the CPU (processor).
2	RAM Slots (DIMM)	Hold RAM modules for temporary data storage.
3	PCIe x16 Slot	Connects a graphics card (GPU).
4	PCIe x1 Slots	Connect expansion cards, such as sound or network cards.
5	SATA Ports	Connect HDDs and SATA SSDs.
6	M.2 Slot	Connects compatible M.2 SSDs.
7	24-Pin ATX Power Connector	Supplies power to the motherboard.

No.	Component Name	Function
8	CPU Power Connector	Supplies power to the CPU.
9	USB Headers	Connect the front USB ports of the computer case.
10	Front Panel Header	Connects the power switch, reset switch, and indicator lights.
11	Rear I/O Ports	Connect external devices such as USB cables, monitors, and keyboards.
12	CMOS Battery	Helps maintain the system clock and BIOS/UEFI settings.

2.

Research and list 3 popular CPU socket types (for example: LGA1151, AM4, LGA1200) and name one compatible CPU model for each.

Three Popular CPU Socket Types and Compatible CPUs

A **CPU socket** is the physical connector on a motherboard where the processor is installed. Different CPU sockets support different processor models, so the socket type must be compatible with the CPU.

S. No.	CPU Socket Type	Compatible CPU Model	Manufacturer
1.	LGA 1151	Intel Core i7-9700K	Intel
2.	AM4	AMD Ryzen 5 5600X	AMD
3.	LGA 1200	Intel Core i7-10700K	Intel

Explanation: These are three popular CPU socket types used in desktop computers. The Intel Core i7-9700K uses the LGA 1151 socket, the AMD Ryzen 5 5600X uses the AM4 socket, and the Intel Core i7-10700K uses the LGA 1200 socket.

3.

Explain in your own words the primary functions of the CPU and the chipset on a motherboard, and describe how they work together when you open a music app like Spotify.

3. Explain the Primary Functions of the CPU and Chipset on a Motherboard and Describe How They Work Together When You Open Spotify.

Answer:

1. Function of the CPU (Central Processing Unit):

The CPU is known as the *brain of the computer*. It executes instructions, performs calculations, processes data, and controls the operations of different programs. When we

open a music application like Spotify, the CPU processes the application's instructions, manages its tasks, and helps the application run smoothly.

2. Function of the Chipset:

The chipset is a group of controllers on the motherboard that manages communication between the CPU and other hardware components. It helps coordinate data transfer between devices such as USB ports, storage devices, and other connected components. The chipset works with the CPU to ensure that different hardware components communicate properly.

3. How the CPU and Chipset Work Together When Opening Spotify:

When we open Spotify, the following steps take place:

1. **Application Loading:** The CPU processes the instructions required to open Spotify, while the operating system loads the application from the SSD or HDD into RAM.
2. **Data Communication:** The chipset helps manage communication between the CPU and supported devices, including storage and USB devices.
3. **Processing:** The CPU executes Spotify's instructions and processes user actions, such as searching for songs or selecting a playlist.
4. **Internet Connection:** The network controller helps connect the computer to the internet so Spotify can retrieve music and other information.
5. **Playing Music:** The CPU processes the application tasks, and the audio hardware converts the digital audio data into sound that we can hear through speakers or headphones.

4.Find a clear photo of a real motherboard (from the internet or your own device), and identify at least 5 different ports or slots by circling them and labeling each one.

The motherboard is the main circuit board of a computer. It connects important components such as the CPU, RAM, storage devices, and graphics card.

Labeled Motherboard Diagram

The following diagram shows a motherboard with its ports and slots circled in red and labeled for easy identification.

Motherboard Components Reference – Tom's Hardware

The diagram above identifies the following components:

1. **CPU Socket:** Holds the Central Processing Unit (CPU).
2. **RAM Slots (DIMM):** Hold RAM modules for temporary data storage.
3. **PCIe x16 Slot:** Connects a graphics card or another compatible expansion card.
4. **SATA Ports:** Connect storage devices such as HDDs and SATA SSDs.
5. **24-pin ATX Power Connector:** Supplies power to the motherboard.
6. **CPU Power Connector:** Supplies power to the processor through the motherboard's power circuitry.
7. **M.2 SSD Slot:** Connects a compatible M.2 storage device.

8. **CMOS Battery:** Helps maintain the system clock and BIOS settings when the computer is powered off.
9. **USB Header:** Connects the computer case's front USB ports to the motherboard.
10. **Rear I/O Ports:** Connect external devices such as a keyboard, mouse, monitor, and network cable.

